

Project Executable Files

Date	03 October 2026
Team ID	SWTID-2026-7859
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	3 Marks

Project Executable Files

All executable and deployable files for EduGenie are in the GitHub repository <https://github.com/Arumparithi42/EduGenie-AI>.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes (README.md)
3	requirements.txt / package.json / dependency file	Yes (requirements.txt, requirements-local.txt)
4	Database schema / seed files (if applicable)	NA (no database)
5	Environment configuration file (.env.example or similar)	Yes (EduGenie/.env.example)
6	Deployed application URL (if hosted)	NA (runs locally)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA (web application)
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes (EduGenie/tests, 40 passing)
10	Demo video or walkthrough (if required)	

Step 2: File / Folder Structure

```

EduGenie-AI/
|-- EduGenie/
|   |-- main.py           # application
|   |-- gemini_client.py  # FastAPI app + API routes
|   |-- explanation_module.py # shared Gemini client (google-genai)
|   |-- qna.py            # concept explanation (LaMini-Flan-T5 / Gemini)
|   |-- quiz_module.py    # question answering
|   |-- summary_module.py # quiz generation + JSON validation
|   |-- learning_path.py  # summarization
|   |-- templates/index.html # learning recommendations
|   |-- static/style.css  # frontend page
|   |-- static/script.js  # styling
|   |-- tests/            # frontend logic
|   |-- requirements.txt   # pytest tests + load_test.py
|   |-- requirements-local.txt # core dependencies
|   |-- .env.example      # optional local model dependencies
|   |-- pytest.ini        # configuration template
|-- Project Documents/    # phase-wise project documents
|-- .vscode/              # VS Code run / debug / test config
`-- README.md             # setup, run and test guide
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Runs locally at http://127.0.0.1:8000 (not hosted)
Login Credentials (Demo)	Not required - EduGenie has no login. A Gemini API key must be set in <code>EduGenie/.env</code> .
Platform / Hosting Provider	Local machine (Uvicorn); AI inference on Google Cloud (Gemini API)
Repository Link	https://github.com/Arumparithi42/EduGenie-AI
Demo Video Link	

Step 4: Run Instructions

1. Install Python 3.10+ (tick "Add Python to PATH") and VS Code with the Python extension.
2. Clone the repository: `git clone https://github.com/Arumparithi42/EduGenie-AI` and open the EduGenie-AI folder in VS Code.
3. Create and activate a virtual environment: `python -m venv .venv` then `.venv\Scripts\Activate.ps1` (Windows) or `source .venv/bin/activate` (macOS/Linux).
4. Install dependencies: `pip install -r EduGenie/requirements.txt`
5. Get a free Gemini API key from <https://aistudio.google.com/app/apikey>, copy `EduGenie/.env.example` to `EduGenie/.env` and set `GEMINI_API_KEY=<your key>`.
6. Start the server: `cd EduGenie` then `uvicorn main:app --reload` (or press F5 -> "EduGenie: Run server" in VS Code).
7. Open <http://127.0.0.1:8000> in a browser (API docs: <http://127.0.0.1:8000/docs>).
8. Run the tests: `pytest -v` (from the EduGenie folder).

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Requires an internet connection and a valid Gemini API key; Gemini models are sometimes overloaded (HTTP 503 "high demand") and free-tier keys are rate-limited (HTTP 429).	Handled automatically: retries with exponential backoff, then backup models (GEMINI_FALLBACK_MODELS). If all are busy, the UI shows a message with a Try again button.
2	The optional local LaMini-Flan-T5 model needs ~3 GB download and the first explanation is slow.	Leave EXPLAIN_BACKEND=auto without installing it, or set EXPLAIN_BACKEND=gemini.
3	AI-generated answers can occasionally be inaccurate.	Prompts ask for accuracy; quiz output is validated; users should verify important facts.